Lección 13

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Plan de trabajo

Teorema de Noether
—Formulación (repaso)
*relación con simetrías
**Resolución de problemas
Formulación lagrangiana de partícula relativista

▶ **Nuestro Syllabus:**

FIS-310:

UNIDAD 3. Teoremas de Conservación
3.1. Teorema de Noether.
3.2. Aplicaciones del teorema de Noether.

UNIDAD 4. Formulación Lagrangiana de la Partícula Relativista
4.1. Lagrangiano de la Partícula Relativista
4.2. Ecuaciones de Lagrange para Partículas Relativista
4.3. Aplicaciones

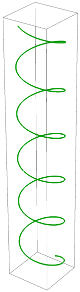
FIS-345:

Unidad I: Principios Variacionales, Lagrangianos y Hamiltonianos
- Principio de Hamilton y Lagrangianos
- Simetrías y Leyes de Conservación (Teorema de Noether)

Unidad II: Algunas Aplicaciones
- Movimiento en campo de Fuerzas Centrales
- Movimiento de Sólido Rígido

Noether theorem and symmetries

We have already seen the Noether's theorem in Lecture 6:



Noether's theorem (combined)

If there is a continuous transformation of coordinates and time

$$\boldsymbol{q} \quad \to \boldsymbol{q} + \delta\boldsymbol{q}(\varepsilon), \qquad t \to t + \delta t(\varepsilon)$$

which keeps the action invariant, than the *Noether invariant*

$$Q = \lim_{\varepsilon\to 0} \frac{\partial L_\varepsilon}{\partial \dot{\varepsilon}}$$

is conserved, $dQ/dt = 0$.

►includes energy, momentum, and angular momentum conservation as special cases

– In mathematical books Q is called *first integral*. (see e.g. *Arnold* from bibliography)
  - * In general $Q = Q(\boldsymbol{q}, \dot{\boldsymbol{q}}, t)$ or equivalently $Q = Q(\boldsymbol{q}, \boldsymbol{p}, t)$

– We discussed basic assumption, with many nice examples, so no need to repeat it again.
  - *We'll focus on less "trivial" questions now

Warm-up questions ...

– The symmetry of action allows us to construct Q. But is it possible to reconstruct symmetry from the invariant $Q(\boldsymbol{q}, \boldsymbol{p}, t)$? Does it "remember" about the underlying symmetry?

– The symmetry transformations of the system always form a group (G):
$\Rightarrow$if $g_1 \in G$ and $g_2 \in G$, their superposition also belongs to the same group $g_1 \cdot g_2 \in G$
$\Rightarrow$Existence of some elements might require existence of others
What does this mean for the invariants $Q_1, Q_2, \ldots$?
How many independent invariants has the system ?

– What happens if the equations of motion are invariant, but the action is NOT? Are there any "extended" invariants associated with this symmetry?
Example (Open question from Lecture 6): Coulomb field, symmetry:

$$\boldsymbol{r} \quad \rightarrow \lambda^2 \boldsymbol{r}, \quad t \rightarrow \lambda^3 t, \tag{1}$$

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Noether theorem and symmetries

- Now we will discuss some additional aspects related to transformation of different observables under continuous symmetry transformations and properties of generators
- We will assume that all quantities are functions of $\boldsymbol{p}$ (generalized momenta) and $\boldsymbol{q}$ (generalized coordinates). This is similar to what is done in Hamiltonian formalism which You've seen earlier in undergraduate courses. We'll see hamiltonian formulation later in more detail
- We will focus only on things relevant for the symmetry

I will follow the Chapter 9.6-9.8 from (*Goldstein*)

► We'll use Poisson's bracket

$$[f,\, g] = \sum_a \left(\frac{\partial f}{\partial p_a} \frac{\partial g}{\partial q_a} - \frac{\partial f}{\partial q_a} \frac{\partial g}{\partial p_a} \right)$$

$$[f,\, q_a] = \frac{\partial f}{\partial p_a}, \quad [f,\, p_a] = -\frac{\partial f}{\partial q_a}$$

► Consider this just as mathematical object. We'll see its physical meaning later, when we'll study hamiltonian dynamics

Assume:

►$(q,\ p) \to (q^{\varepsilon},\ p^{\varepsilon})$, is a continuous symmetry transformation of the action
 – ε is a parameter (e.g rotation angle or similar; $\varepsilon \ll 1$ for simplicity)

► $Q(\boldsymbol{p},\ \boldsymbol{q})$ is a Noether invariant (integral of motion) corresponding to this symmetry
 – $Q(\boldsymbol{p},\ \boldsymbol{q})$ does not depend on ε explicitly.

►For simplicity consider 1D case
–Noether invariant construction from the Noether's theorem:

$$Q(\boldsymbol{p},\ \boldsymbol{q}) \qquad = \gamma \cdot \frac{\partial L}{\partial \dot{q}} = \frac{\partial q}{\partial \varepsilon} \cdot p, \quad p \equiv \frac{\partial L}{\partial q}, \qquad \frac{\partial Q}{\partial p} = \gamma = \frac{\partial q^{\varepsilon}}{\partial \varepsilon}$$

► Explicit change of Q under transformation $(q,\ p) \to (q^{\varepsilon},\ p^{\varepsilon})$

$$\begin{aligned} 0 = \frac{dQ}{d\varepsilon} \qquad &= \frac{\partial Q}{\partial p}\,\frac{\partial p^{\varepsilon}}{\partial \varepsilon} + \frac{\partial Q}{\partial q}\,\frac{\partial q^{\varepsilon}}{\partial \varepsilon} \\ \Rightarrow \frac{\partial p^{\varepsilon}}{\partial \varepsilon} \qquad &= -\frac{\partial Q}{\partial q}\,\cancel{\frac{\partial q^{\varepsilon}/\partial \varepsilon}{\partial Q/\partial p}} \end{aligned}$$

- Assume $G(\boldsymbol{p},\ \boldsymbol{q})$ is ***arbitrary*** function of momenta $\boldsymbol{p}$ and coordinates $\boldsymbol{q}$

– then its change under symmetry transformations

$$\begin{aligned} \frac{dG}{d\varepsilon} \quad &= \frac{\partial G}{\partial \varepsilon} + \frac{\partial G}{\partial p}\frac{\partial p^{\varepsilon}}{\partial \varepsilon} + \frac{\partial G}{\partial q}\frac{\partial q^{\varepsilon}}{\partial \varepsilon} = \frac{\partial G}{\partial \varepsilon} - \frac{\partial G}{\partial p}\frac{\partial Q}{\partial q} + \frac{\partial G}{\partial q}\frac{\partial Q}{\partial p} \\ &= \frac{\partial G}{\partial \varepsilon} + [Q,\ G] := \frac{\partial G}{\partial \varepsilon} + \hat{\boldsymbol{Q}}\, G \end{aligned}$$

- $\hat{\boldsymbol{Q}}$ is called generator, and $[Q,\ G]$ is Poisson bracket

$\Rightarrow Q(\boldsymbol{p}, \boldsymbol{q})$ is not just constant along trajectory, it also tells how other quantities change under this symmetry

–Frequently G does not depend on ε, so its change is given by

$$\delta G(\boldsymbol{p},\ \boldsymbol{q}) \qquad = \varepsilon\, \hat{\boldsymbol{Q}} G \equiv \varepsilon [Q,\ G(\boldsymbol{p},\ \boldsymbol{q})]$$

Examples

- Assume that symmetry of action corresponds to translations $\boldsymbol{r} \to \boldsymbol{r} + a\,\boldsymbol{n}$

$-$Noether invariant $Q = \boldsymbol{p} \cdot \boldsymbol{n}$

$-$Change of arbitrary function under translations: $G(\boldsymbol{r}) \to G(\boldsymbol{r}) + \delta G(\boldsymbol{r})$,

$$\delta G(\boldsymbol{r}) \qquad = a\,\boldsymbol{n} \cdot \frac{\partial G(\boldsymbol{r})}{\partial \boldsymbol{r}} = a\,\boldsymbol{n} \cdot [\boldsymbol{p},\, G(\boldsymbol{r})] = a\,[\boldsymbol{p} \cdot \boldsymbol{n},\, G(\boldsymbol{r})]$$

- Similarly for rotations around axis $\boldsymbol{n}$ (angle $\phi \ll 1$):

$$\boldsymbol{r} \to \boldsymbol{r} + \phi \boldsymbol{n} \times \boldsymbol{r}, \qquad \boldsymbol{p} \to \boldsymbol{p} + \phi \boldsymbol{n} \times \boldsymbol{p}$$

$$\begin{aligned}
\delta G(\boldsymbol{p},\, \boldsymbol{r}) \qquad &= \phi\,(\boldsymbol{n} \times \boldsymbol{p}) \cdot \frac{\partial G}{\partial \boldsymbol{p}} + \phi\,(\boldsymbol{n} \times \boldsymbol{r}) \cdot \frac{\partial G}{\partial \boldsymbol{r}} \\
&= \phi\,(\boldsymbol{n} \times \boldsymbol{p}) \cdot [G,\, \boldsymbol{r}] - \phi\,(\boldsymbol{n} \times \boldsymbol{r}) \cdot [G,\, \boldsymbol{p}] \\
&= \phi\,[G,\, \boldsymbol{n} \cdot (\boldsymbol{p} \times \boldsymbol{r})] = \phi\,[\boldsymbol{M} \cdot \boldsymbol{n},\, G(\boldsymbol{p},\, \boldsymbol{r})]
\end{aligned}$$

► $\boldsymbol{M} \cdot \boldsymbol{n}$, $\boldsymbol{p} \cdot \boldsymbol{n}$ $\leftrightarrow$ *generators* of evolution, rotations, translations

$$dG/da = [\boldsymbol{p} \cdot \boldsymbol{n},\, G], \qquad dG/d\phi = [\boldsymbol{M} \cdot \boldsymbol{n},\, G]$$

Finite transformations

If we know generator, can reconstruct the change of function for *finite* transformations (*Goldstein*, §9.6):

▷ Generalization for arbitrary transformation $\boldsymbol{q} \to \tilde{\boldsymbol{q}}(\epsilon)$:

$$\begin{aligned} \tilde{\boldsymbol{q}}(\epsilon) &= \boldsymbol{q} + \epsilon \hat{\boldsymbol{Q}}\, \boldsymbol{q} + \frac{\epsilon^2}{2!} \hat{\boldsymbol{Q}}^2 \boldsymbol{q} + \ldots = \exp\left(\epsilon \hat{\boldsymbol{Q}}\right) \boldsymbol{q} \\ G\left(\tilde{\boldsymbol{q}}(\epsilon)\right) &= G(\boldsymbol{q}) + \epsilon \hat{\boldsymbol{Q}}\, G(\boldsymbol{q}) + \frac{\epsilon^2}{2!} \hat{\boldsymbol{Q}}^2 G(\boldsymbol{q}) + \ldots = \exp\left(\epsilon \hat{\boldsymbol{Q}}\right) G(\boldsymbol{q}) \end{aligned}$$

$\Rightarrow$ $\hat{\boldsymbol{Q}}$ (generator) fully characterizes any continuous transformation.

* *Beware the difference in the literature: we use definition as it appears in Classical Mechanics and Mathematics textbooks. In Quantum Mechanics* $\hat{\boldsymbol{Q}}$ *includes additional factor "i" in definition, so* $\tilde{\boldsymbol{q}} = \exp\left(i\epsilon \hat{\boldsymbol{Q}}\right) q \approx \boldsymbol{q} + i\,\epsilon \hat{\boldsymbol{Q}}\, \boldsymbol{q}$.

Noether theorem and generators*

Our goal: Illustrate relation between integrals of motion in classical Mechanics and generators of symmetry group of the action

- We'll define briefly Lie group and its algebra (see e.g. Arfken, Weber, *Mathematical methods for Physicists*, chapter 17 for more details)
- We'll discuss relation between generators of symmetries and integrals of motion (*Goldstein*, §9.8 for more details). This would let us use very powerful tools from theory of Lie groups for analysis

Main result (informal): *Sometimes existence of certain symmetry elements requires existence of other symmetry elements (and associated Noether integrals of motion)*

Noether theorem and generators*

- If we have a 1-parametric symmetry transform

$$(\boldsymbol{q}, \boldsymbol{p}) \to (\boldsymbol{q}^{\varepsilon}, \boldsymbol{p}^{\varepsilon}) \quad (1)$$

which keeps the action invariant, and ε is some continuous parameter, then (1) is *canonical* transformation.

- All possible (continuous) symmetry transformations T which keep the action invariant and satisfy group axioms (see right) form its symmetry group

Reminder: Group axioms

▷Closeness with respect to superposition (multiplication):
If $T_1 \in \boldsymbol{T}$, $T_2 \in \boldsymbol{T}$, then $\exists T_3 \in \boldsymbol{T}$, $T_3 = T_1 \cdot T_2$

▷Existence of unity (identical transform $E \equiv \hat{1}$)

$$\forall T_1 \in \boldsymbol{T}, \quad T_1 \cdot E = E \cdot T_1 = T_1$$

▷Existence of inverse: for $\forall T_1$ $\exists! T_1^{-1} \in \boldsymbol{T}$, $\Rightarrow T_1 \cdot T_1^{-1} = T_1^{-1} T_1 = E$

▷Associativity:

$$(T_1 \cdot T_2) \cdot T_3 = T_1 \cdot (T_2 \cdot T_3)$$

▷ If $T_1 T_2 = T_2 T_1$ for $\forall T_1, T_2$, the group is called abelian (commutative), otherwise nonabelian.

e.g. all translations form abelian group; all rotations form nonabelian group.

▷Group is continuous (Lie) if we can introduce "coordinates" ("map all elements to $\mathbb{R}^n$", "construct isomorphism to $\mathbb{R}^n$").

Noether theorem and generators*

►Let's assume that transformations are close to unit element,

$$T_1 \approx \hat{1} + \alpha_1 \hat{\boldsymbol{Q}}_1, \; T_2 \approx \hat{1} + \alpha_2 \hat{\boldsymbol{Q}}_2$$

(e.g. T_1 and T_2 are rotations around $\hat{x}$ and $\hat{y}$)

▷ Let's evaluate "commutator"

$$\begin{aligned} T_{12} &= T_1 T_2 T_1^{-1} T_2^{-1} \approx \\ &\approx \hat{1} + \alpha_1 \alpha_2 \left(\hat{\boldsymbol{Q}}_1 \hat{\boldsymbol{Q}}_2 - \hat{\boldsymbol{Q}}_2 \hat{\boldsymbol{Q}}_1 \right) \end{aligned}$$

(terms $\sim \alpha_1, \sim \alpha_2, \sim \alpha_1^2, \sim \alpha_2^2$ cancel)

▷ T_{12} is a superposition, so it MUST have a form

$$T_{12} \approx \hat{1} + \sum \beta_k \boldsymbol{Q}_k$$

$$\Rightarrow \hat{\boldsymbol{Q}}_a \hat{\boldsymbol{Q}}_b - \hat{\boldsymbol{Q}}_b \hat{\boldsymbol{Q}}_a \equiv \left[\hat{\boldsymbol{Q}}_a, \, \hat{\boldsymbol{Q}}_b \right] = \sum_k f_{ab}^k \hat{\boldsymbol{Q}}_k$$

-commutator of generators MUST be a linear combination of other generators. f_{ab}^k are called structure constants and fully characterize the Lie group.

►Mathematical language: *The generators form a Lie algebra, closed with respect to commutators of any two elements*

Noether theorem and generators*

►Earlier we used notation $[f,\, g] \equiv [f,\, g]_{p,q}$ for Poisson bracket

▷We defined operator $\hat{\boldsymbol{f}}$ and $\hat{\boldsymbol{g}}$ such that when we act with them on *any* function h,

$$\hat{\boldsymbol{f}}\, h \equiv [f,\, h]_{p,q}, \quad \hat{\boldsymbol{g}}\, h \equiv [g,\, h]_{p,q},$$

Now we use the notation $\left[\hat{\boldsymbol{f}},\, \hat{\boldsymbol{g}}\right]$ for commutator of operators (also operator).

Lemma

$$\left[\hat{\boldsymbol{f}},\, \hat{\boldsymbol{g}}\right] h \equiv [[f,\, g]_{p,q},\, h]_{p,q},$$

i.e. Poisson bracket of $[f,\, g]_{p,q}$ corresponds to generator $\left[\hat{\boldsymbol{f}},\, \hat{\boldsymbol{g}}\right]$

►Proof:

$$\begin{aligned}
\left[\hat{\boldsymbol{f}},\, \hat{\boldsymbol{g}}\right] h &\equiv [f,\, [g,\, h]_{p,q}]_{p,q} - [g,\, [f,\, h]_{p,q}]_{p,q} \\
&= [f,\, [g,\, h]_{p,q}]_{p,q} + [g,\, [h,\, f]_{p,q}]_{p,q} \\
&= -[h,\, [f,\, g]_{p,q}]_{p,q} = [[f,\, g]_{p,q},\, h]_{p,q} \qquad \text{Q.E.D.}
\end{aligned}$$

($2^{\text{nd}} \to 3^{\text{rd}}$ line: use Jacobi identity)

Noether theorem and generators*

- *Mathematical point of view*:

The *algebra of generators should be closed*, *i.e.* commutator of 2 generators is a linear combination of other generators.

$\Rightarrow$If $f(\boldsymbol{p},\, \boldsymbol{q})$ and $g(\boldsymbol{p},\, \boldsymbol{q})$ are generators of some symmetry, then Poisson bracket $[f,\, g]_{p,q}$ is also a generator of the symmetry (integral of motion)

$-$Correspondence *integral of motion* $\leftrightarrow$ *generator* allows to use powerful apparatus of Lie groups in Classical Mechanics

$-$We will postpone the discussion until we will learn the Hamiltonian formulation of Classical Mechanics

Example 1

The hamiltonian of a particle is invariant w.r.t. rotations around $\hat{x}$ and translations in direction $\hat{y}$. Prove that the action of the system should be also independent of invariant w.r.t. translations in direction of $\hat{z}$ (i.e. the potential U may depend only on coordinate x).

- The generators of the symmetries mentioned in the problem are angular momentum component M_x and momentum p_y
- We will try to evaluate the commutators sequentially until we get closed set:

$$
\begin{aligned}
[M_x, p_y] &= -p_z, \\
[M_x, p_z] &= p_y, \\
[p_y, p_z] &= 0
\end{aligned}
$$

Thus we got closed algebra: pairwise commutators of M_x, p_y, p_z give the elements from the same set.

– The symmetry group includes rotation around x ($\varphi_x \to \varphi_x + \mathrm{const}$) and two translations ($y \to y + \mathrm{const}$ and $z \to z + \mathrm{const}$)

Example 2

The hamiltonian of a particle is invariant w.r.t. rotations around $\hat{x}$ and $\hat{y}$. Demonstrate that it is invariant w.r.t. rotation around any axis $\boldsymbol{n}$, i.e. potential $U(\boldsymbol{r}) = U(r)$ (central field).

►The generators of the symmetries mentioned in the problem are angular momentum components M_x and M_y

►We will try to evaluate the commutators sequentially until we get closed set:

$$
\begin{aligned}
[M_x, M_y] &= -M_z, \\
[M_x, M_z] &= -M_y, \\
[M_y, M_z] &= -M_x,
\end{aligned}
$$

For rotation around $\boldsymbol{n}$ generator $Q \equiv \boldsymbol{M} \cdot \boldsymbol{n} = M_x n_x + M_y n_y + M_z n_z$ and thus also belongs to symmetry group of the hamiltonian.

Example 3

In Lecture 6 we discussed that for a free particle the action is invariant with respect to rescaling transformation

$$\boldsymbol{r} \to \alpha\, \boldsymbol{r}, \;\; t \to \alpha^2 t.$$

and the associated Noether invariant $Q = \boldsymbol{p} \cdot \boldsymbol{r} - 2E\, t$. Identify other symmetry elements which has the action of this system and write out the corresponding Noether invariants. Using the Poissons's theorem, demonstrate that they form a closed algebra, and identify symmetry group of the action.

Example 3 (solution)

–System is invariant w.r.t. rotations, translations (time, space) and rescaling

The Noether invariants in this system are $Q = \boldsymbol{p} \cdot \boldsymbol{r} - 2E\,t$, hamiltonian $H = p^2/2m$ ("energy"), components p_a of the momentum, and components M_a of the angular momentum. Clearly,

$$[p_a, p_b] = [p_a, H] = [M_a, H] = 0,$$

$$[M_a, M_b] = -\epsilon_{abc} M_c, \;\; [M_a, p_b] = -\epsilon_{abc} p_c,$$

$$[Q,\; p_a] = [\boldsymbol{p} \cdot \boldsymbol{r} - 2H\,t,\; p_a] = -p_a$$

$$[Q,\; M_a] = [\boldsymbol{p} \cdot \boldsymbol{r} - 2H\,t,\; M_a] = 0$$

$$[Q,\; H] = \sum_a [Q,\; p_a]\,\frac{p_a}{m} = -\sum_a \frac{p_a^2}{m} = -2H$$

–Further application of the Poisson's theorem does not give any new invariants (algebra is closed)

–The group which we found is a subgroup of the so-called Galilean conformal group (the latter also includes boosts).

Example 4

In Lecture 6 we discussed that for a particle moving in the central field $U(r) = -\alpha/r^2$ the action is invariant with respect to rescaling transformation

$$\boldsymbol{r} \to \alpha\, \boldsymbol{r}, \;\; t \to \alpha^2 t.$$

and the associated Noether invariant $Q = \boldsymbol{p} \cdot \boldsymbol{r} - 2E\, t$. Identify other symmetry elements which has the action of this system and write out the corresponding Noether invariants. Using the Poissons's theorem, demonstrate that they form a closed algebra, and identify symmetry group of the action.

Example 4 (solution)

–System is invariant w.r.t. rotations, translation $t \to t + t_0$ and rescaling
The Noether invariants in this system are $Q = \boldsymbol{p} \cdot \boldsymbol{r} - 2E\,t$, hamiltonian $H = p^2/2m + U$ ("energy"), and components M_a of the angular momentum. Clearly,

$$[M_a, M_b] = -\epsilon_{abc} M_c, \;\; [M_a, H] = 0,$$

$$[Q,\, M_a] = [\boldsymbol{p} \cdot \boldsymbol{r} - 2H\,t,\, M_a] = 0$$

$$[Q,\, H] = \sum_a [Q,\, p_a]\,\frac{p_a}{m} + [Q,\, U] = -\sum_a \frac{p_a^2}{m} - r_a \underbrace{[p_a,\, U]}_{\partial U/\partial r^a}$$

$$= -\sum_a \frac{p_a^2}{m} - r_a \frac{\partial U}{\partial r^a} = -\sum_a \frac{p_a^2}{m} - 2U = -2H$$

–Further application of the Poisson's theorem does not give any new invariants.
–The group which we found is a subgroup of what we found in the previous problem (no translations now)

Open question

Assume that the **equations of motion** are invariant under a continuous symmetry transformation

$$\boldsymbol{r} \to \boldsymbol{r}' = \boldsymbol{r}'(\lambda, \boldsymbol{r}), \quad t \to t' = t'(\lambda, t)$$

where λ is a continuous parameter, but the action is NOT. Does this imply that there is a Noether invariant ?

- Clearly, if the action is invariant under some symmetry, then the equations of motion are also invariant.
- However, we have seen that the same equation can be found from different lagrangians, so for some symmetries **the equations are invariant,** but the action is NOT: we'll see **example in the next page**

Example - symmetry of EOMs which changes action

A particle moves in the Coulomb field, $U(r) = \alpha/r$, transformation

$$\boldsymbol{r} \to \lambda^2 \boldsymbol{r}, \quad t \to \lambda^3 t, \tag{2}$$

Equation of motion (EOM):

$$m\frac{d^2\boldsymbol{r}}{dt^2} = \boldsymbol{F} = \frac{\alpha}{r^2}\hat{\boldsymbol{r}} \tag{3}$$

−Transformation of different quantities:

$$\frac{d\boldsymbol{r}}{dt} \to \frac{1}{\lambda}\,\frac{d\boldsymbol{r}}{dt},$$

$$m\frac{d^2\boldsymbol{r}}{dt^2} \to \frac{1}{\lambda^4}\, m\frac{d^2\boldsymbol{r}}{dt^2}, \qquad \boldsymbol{F} = \frac{\alpha}{r^2}\hat{\boldsymbol{r}} \to \frac{1}{\lambda^4}\,\frac{\alpha}{r^2}\hat{\boldsymbol{r}} \Rightarrow \text{EOM (2) is invariant}$$

−Transformation of lagrangian and action:

$$T = m\left(\frac{dr}{dt}\right)^2 \quad \to \frac{m}{\lambda^2}\left(\frac{dr}{dt}\right)^2, \qquad U(r) = \frac{\alpha}{r} \to \frac{1}{\lambda^2}\,\frac{\alpha}{r}, \qquad L = T - U \to \frac{1}{\lambda^2}\,L$$

$$S = \int dt\, L \to \lambda S \neq S \Rightarrow \text{Action is not invariant}$$

Please let me know what do You think:

−Are there any Noether invariants associated with this symmetry ?

−What we would get if we try to derive "invariant" Q using conventional procedure?

Derivation of the "invariant"

Standard procedure for construction of Noether "invariants" :
$-$assume $\lambda = e^{\varepsilon(t)}$ where $|\varepsilon| \ll 1$.
$-$Need to focus on $\dot{\varepsilon}$-terms (terms $\sim \varepsilon$ vanish in view of the symmetry of the action).
In lagrangian such terms come only from the kinetic energy:

$$\frac{d\boldsymbol{r}}{dt} \to \frac{d\left(\lambda^2 \boldsymbol{r}\right)/dt}{d\left(\lambda^3 t\right)/dt} \approx \frac{\boldsymbol{v} + 2\dot{\varepsilon}\boldsymbol{r}}{1 + 3\,\dot{\varepsilon} t} + \mathcal{O}\left(\varepsilon,\, \varepsilon^2,\, \varepsilon\dot{\varepsilon} \ldots\right) \approx \boldsymbol{v} + \dot{\varepsilon}\left(2\boldsymbol{r} - 3\,t\boldsymbol{v}\right)$$

Change of lagrangian:

$$L \to L + m\boldsymbol{v}\cdot\left(2\boldsymbol{r} - 3\,t\boldsymbol{v}\right)\dot{\varepsilon} + \mathcal{O}\left(\varepsilon,\, \varepsilon^2,\, \varepsilon\dot{\varepsilon}\ldots\right)$$

$-$Should take into account transformation of time differential in $S = \int L dt$

$$dt \to dt\left(1 + \frac{\partial \lambda^3}{\partial t} t\right) \approx dt\left(1 + 3\dot{\varepsilon} t\right)$$

$$Q = \frac{\delta S}{\delta \dot{\varepsilon}} \qquad = m\boldsymbol{v}\cdot\left(2\boldsymbol{r} - 3\,t\boldsymbol{v}\right) + 3Lt = m\boldsymbol{v}\cdot\left(2\boldsymbol{r} - \frac{3}{2}\,t\boldsymbol{v}\right) + 3\frac{\alpha}{r} =$$

$$= 2m\boldsymbol{v}\cdot\boldsymbol{r} - 3Et = 2\boldsymbol{p}\cdot\boldsymbol{r} - 3Et$$

$$\frac{dQ}{dt} \qquad = 2\frac{d\boldsymbol{p}}{dt}\cdot\boldsymbol{r} + 2\boldsymbol{p}\cdot\frac{d\boldsymbol{r}}{dt} - 3E = 2\left(-\frac{\partial U}{\partial \boldsymbol{r}}\right)\cdot\boldsymbol{r} + 2\boldsymbol{p}\cdot\frac{d\boldsymbol{r}}{dt} - 3E$$

$$= -\frac{2\alpha}{r} + 2mv^2 - 3\frac{mv^2}{2} + \frac{3\alpha}{r} = \left(\frac{mv^2}{2} + \frac{\alpha}{r}\right) = +L \neq 0 \text{ (?!)}$$

Analysis

−I will follow analysis of [J. Phys. A: Math. Gen. 14 (1981) 587-596] (later reprinted in many textbooks).

Sophus Lie: there are invariants associated with the symmetry of a system of ODEs.

$$\dot{x}_a = f_a\left(x_1, ..., x_n,\ t\right)$$

which are invariant under 1-parametric transformation of coord. and time: If s is some parameter and $x_a\left(t, s = 0\right)$ is solution, then $x_a(t, s)$ must be also solution (provided initial conditions are properly adjusted).

−ODEs can be Euler-Lagrange equations, but we do not require invariance of the action which correspond to them now

Examples of symmetries:

− Invariance $t \to t + t_0$: if $x(t)$ is solution, then $x(t + t_0)$ is also a solution

$$\begin{aligned} x_a\left(t + t_0\right) &= \underbrace{\left[1 + t_0 \frac{d}{dt} + \frac{t_0^2}{2}\frac{d^2}{dt^2} + ...\right] x_a\left(t\right)}_{\text{Taylor expansion}} \\ &= \exp\left[t_0 \frac{d}{dt}\right] x_a\left(t\right) \end{aligned}$$

−Rotation invariance $\phi \to \phi + \phi_0$: as in previous case, just replace $t, t_0, dt \to \phi, \phi_0, d\phi$

Analysis

Generalization: consider symmetry

$$t \to t + s_0\xi, \qquad x_a \to x_a + s_0\eta_a$$

In general ξ, $\eta_a$are functions of (t, x)

$$x_a(t, s_0) \qquad = \left[1 + s_0\frac{d}{ds} + \frac{s_0^2}{2}\frac{d^2}{ds^2} + \ldots\right] x_a(t) = \exp\left[s_0\frac{d}{ds}\right] x_a(t)$$

–We'll call the the vector

$$\hat{X} \equiv \frac{d}{ds} = \xi\frac{\partial}{\partial t} + \eta_a\frac{\partial}{\partial x^a}$$

a generator of the symmetry group.

–Can check that for time shifts, translations and rotations, operator $\hat{X}$ corresponds to $\partial/\partial t$, $\partial/\partial x_a$ and $\varepsilon_{abc}x_b\partial_c$ respectively (compare with operators of linear and angular momentum in QM).

–A function $F(x_a, t)$ is *invariant*, if under transformations of symmetry group

$$F(x_a(t, s), t, s) = F(x_a(t, 0), t, 0)$$

Expanding over s, we may get

$$\hat{X}F = 0 \qquad (1)$$

(clearly (1) is necessary and sufficient condition to have invariant surface)

Analysis

– The operator $Q = 2\boldsymbol{p} \cdot \boldsymbol{r} - 3Et$ found earlier corresponds to generator

$$\hat{X} = \hat{X}_3 = t\frac{\partial}{\partial t} - \frac{2}{3}\left(\boldsymbol{r} \cdot \boldsymbol{\nabla}\right)$$

of some hidden symmetry group, with $\xi \equiv t$, $\eta_a = -\frac{2}{3}x_a$.
Other invariants for 2D Kepler problem:

$$\hat{X}_1 = \partial/\partial t, \qquad \hat{X}_2 = x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}$$

Corresponding Lie algebra:

$$[X_1, X_2] = 0, \qquad [X_1, X_3] = X_1, \qquad [X_2, X_3] = 0,$$

The solutions of

$$\hat{X}_3 F = 0 \qquad (1)$$

can be found using the method of characteristics, solving

$$\frac{dt}{\xi} = \frac{dx}{\eta_x} = \frac{dy}{\eta_y} = \frac{dz}{\eta_z} = ds$$

To simplify analysis, we'll consider only motion in a plane Oxy (2D)

Analysis

– Characteristics:

$$U_1 = xt^{-2/3}, \quad U_2 = yt^{-2/3}, \qquad V_1 = \dot{x}t^{1/3}, \quad V_2 = \dot{y}t^{1/3}$$

Note that we have $X_1 = \partial/\partial t$ in the list of our symmetry generators (conservative system), so we would like to express everything in a time-independent way, in terms of $x, y, \dot{x}, \dot{y}$. We'll treat U_1 as "independent" variable and others as dependent,

$$\frac{dU_2}{dU_1} = \frac{dU_2/dt}{dU_1/dt} = \frac{\dot{y}t^{-2/3} - \frac{2}{3}yt^{-5/3}}{\dot{x}t^{-2/3} - \frac{2}{3}xt^{-5/3}} = \frac{V_2 - \frac{2}{3}U_2}{V_1 - \frac{2}{3}U_1}$$

$$\frac{dV_1}{dU_1} = \frac{dV_1/dt}{dU_1/dt} = \frac{\ddot{x}t^{1/3} + \frac{1}{3}\dot{x}t^{-2/3}}{\dot{x}t^{-2/3} - \frac{2}{3}xt^{-5/3}} = \frac{-\mu P^{-3}U_1 + \frac{1}{3}V_1}{V_1 - \frac{2}{3}U_1}$$

$$\frac{dV_2}{dU_1} = \frac{dV_2/dt}{dU_1/dt} = \frac{\ddot{y}t^{1/3} + \frac{1}{3}\dot{y}t^{-2/3}}{\dot{x}t^{-2/3} - \frac{2}{3}xt^{-5/3}} = \frac{-\mu P^{-3}U_2 + \frac{1}{3}V_2}{V_1 - \frac{2}{3}U_1}$$

The solution of this system gives an invariant surface $F(U_1, U_2, V_1, V_2) = \text{const}$ ("first integral")

$$F_1 = U_1V_2^2 - U_2V_1V_2 - \mu U_1/P = \text{const}, \quad F_2 = U_2V_1^2 - U_1V_1V_2 - \mu U_2/P = \text{const},$$

$$\frac{dF_a}{dU_1} = \frac{\partial F_a}{\partial U_1} + \frac{\partial F_a}{\partial U_2}\frac{dU_2}{dU_1} + \frac{\partial F_a}{\partial V_1}\frac{dV_1}{dU_1} + \frac{\partial F_a}{\partial V_2}\frac{dV_2}{dU_1} = 0, \qquad a = 1, 2$$

Analysis

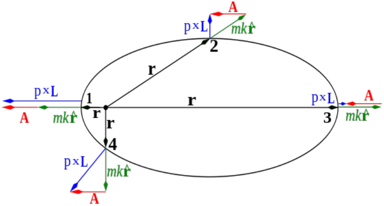
The invariant surfaces in conventional variables:

$$\begin{aligned} x\dot{y}^2 - y\dot{x}\dot{y} - \mu x/r &= \text{const}, \\ y\dot{x}^2 - x\dot{x}\dot{y} - \mu y/r &= \text{const}, \end{aligned}$$

which are just x, y components of Laplace-Runge-Lentz vector

$$\boldsymbol{A} = \boldsymbol{v} \times \boldsymbol{L} - \alpha\,\hat{\boldsymbol{r}}$$

►Orbit keeps its orientation:



►Conservation of Laplace-Runge-Lenz (LRL) vector:

$$\boldsymbol{A} = \boldsymbol{v} \times \boldsymbol{L} - \alpha\,\hat{\boldsymbol{r}}, \qquad A^2 = L^2\frac{2E}{m} + \alpha^2$$

– The absolute value A^2 is expressed via other invariants (E, L), so it is NOT a new invariant. But now we found that its <u>orientation</u> is conserved.

– Since $\boldsymbol{A}\cdot\boldsymbol{L} = 0$, the vector $\boldsymbol{A}$ always stays in the scattering plane $\Rightarrow$ we need just one angle (new independent integral of motion) to characterize orientation of $\boldsymbol{A}$.

Summary

1. The continuous symmetry always implies existence of first integrals (Noether invariants)
2. The Noether invariants $Q(\boldsymbol{p}, \boldsymbol{q})$ are not just conserved "numbers", they also play the role of the generators of the corresponding symmetry
3. The symmetry of the equations of motion implies existence of invariants (first integrals) even if the action S is NOT invariant, however derivation of the first integral is much more sophisticated

Questions & Answers (Q&A)

Please let me know if You have any questions or comments ...

Our next topic:

UNIDAD 4. Formulación Lagrangiana de la Partícula Relativista

4.1. Lagrangiano de la Partícula Relativista

4.2. Ecuaciones de Lagrange para Partículas Relativista

4.3. Aplicaciones

Lagrangian formulation for relativistic particle

▶We considered Lagrangians of nonrelativistic systems, $L = T - U$
−The formalism is quite general, can try to adapt it for relativistic particles
▶We'll start with free particles and later discuss how they interact

Landau: Minimal action for a free particle:

$$S = \int dt\, L = \min$$

where action S
−Must be Lorentz- and rotation covariant,
−Must be translation-invariant
(Group of transformations: Lorentz + Rotations + Translations = Poincaré group)
−For nonrelativistic case should reduce to $S_{\text{nonrel}} = \int dt\ \left(m\, v^2/2\right)$

Please suggest how we could construct the action for a free relativistic particle
−Try to guess or recall if You have seen this in undergraduate courses ...
−Analyze if there are any Lorentz-invariant functionals which have desired properties ...

Lagrangian formulation for relativistic particle

►The only variable which has all desired properties is the particle's proper time τ:

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}$$

►Lagrangian:

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}}$$

−Nonrelativsitic limit:

$$L \approx -mc^2 + \frac{mv^2}{2} + \underbrace{\frac{mv^4}{8c^2}}_{\mathcal{O}(v^2/c^2)} + \ldots$$

Identify the symmetries of the lagrangian and the corresponding Noether invariants

Lagrangian formulation for relativistic particle

►The only variable which has all desired properties is the particle's proper time τ:

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}$$

►Conserved quantities:
– energy E (*consequence of invariance w.r.t.* $t \to t + t_0$)

– momentum $\boldsymbol{p} = \partial L/\partial \boldsymbol{v} = m\boldsymbol{v}/\sqrt{1 - v^2/c^2}$ (*consequence of translational invariance,* $\boldsymbol{r} \to \boldsymbol{r} + \boldsymbol{r}_0$)

–angular momentum (projection to any axis) $M_i = \partial L/\partial \dot{\phi}_i$ (*consequence of invariance w.r.t. rotation around any axis*)

►Euler-Lagrange equation:

$$d\boldsymbol{p}/dt = 0$$

–momentum conservation

Covariant lagrangian

[Sec. 7.10 in Goldstein]

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}, \qquad L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}}$$

Note:

- The action S is Lorentz-invariant, however L is NOT
  $-$(recall how intervals dt change)
- The Euler-Lagrange equations which we found apparently are NOT Lorentz-covariant (include d/dt...)
- The problem started when we switched from $\int d\tau$ to $\int dt$ in action S

Covariant lagrangian

Section 7.10 in Goldstein

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}, \qquad L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}}$$

▶Let's try to keep invariance and NOT to separate out the time and coordinates

– Assume θ is some parameter along the trajectory $x^\mu(\theta)$

– It could be a proper time τ, but not necessarily

$$d\tau = \sqrt{d\tau^2} = \sqrt{\eta_{\mu\nu} dx^\mu dx^\nu} = \sqrt{\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}\, d\theta, \qquad \dot{x}^\alpha \equiv \frac{dx^\alpha}{d\theta}$$

$$S = -mc^2 \int d\tau = -mc^2 \int d\theta\, \Lambda\left(x^\mu,\ \dot{x}^\mu\right), \qquad \Lambda = -mc^2 \sqrt{\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} = -mc^2 \frac{d\tau}{d\theta}$$

–For free particles x^μ-dependence does not appear (translation invariance)

–Corresponding covariant Euler-Lagrange equations:

$$\frac{\partial \Lambda}{\partial x^\mu} - \frac{d}{d\theta}\left(\frac{\partial \Lambda}{\partial\left(\dot{x}^\mu\right)}\right) = 0$$

*For free particle

$$\frac{\partial \Lambda}{\partial\left(\dot{x}^\mu\right)} = -\frac{mc^2 \dot{x}_\mu}{d\tau / d\theta} = -mc^2 u_\mu$$

Motion in external field

- We can use lagrangian formalism for relativistic motion of particles in ***external*** fields
- Main idea: add interaction terms to lagrangian/action
  $-$Should maintain Lorentz covariance of action *S*

  $-$Should be $local$: if $x^\mu(\tau)$ is trajectory, then only values of fields at the point $x^\mu(\tau)$ are relevant.
- Minimal action formalism allows to rewrite the result in covariant form, easier for analysis of Lorentz covariance

Example

1-dimensional motion of a relativistic particle in static 1D potential in lab frame:

$$S = \int dt\, L, \qquad L = -mc^2\sqrt{1 - \frac{v^2}{c^2}} - U(x)$$

$\Rightarrow$ Lorentz covariance explicitly broken since potential is defined in a given frame
– In any other frame $U = U(x, t)$ and interaction term

$$S_{\text{int}} = \int dt\, U(x) = \int d\tau\, \frac{U(x)}{\sqrt{1 - v^2}}$$

- Energy conservation:

$$E = \frac{mc^2}{\sqrt{1 - v^2/c^2}} + U(x) \Rightarrow \frac{dx}{dt} \equiv v = c\sqrt{1 - \left(\frac{mc^2}{E - U(x)}\right)^2}$$

$$\Rightarrow t - t_0 = \frac{1}{c}\int \frac{dx}{\sqrt{1 - \left(\frac{mc^2}{E - U(x)}\right)^2}}$$

Example

$$E \quad = \frac{mc^2}{\sqrt{1-v^2/c^2}} + U(x) \Rightarrow \frac{dx}{dt} \equiv v = c\sqrt{1-\left(\frac{mc^2}{E-U(x)}\right)^2}$$

$$\Rightarrow t - t_0 = \frac{1}{c}\int \frac{dx}{\sqrt{1-\left(\frac{mc^2}{E-U(x)}\right)^2}}$$

▶**Nonrelativistic limit**: $E = mc^2 + \mathcal{E}, \quad \mathcal{E} \ll mc^2$

$$\Rightarrow t - t_0 \approx \frac{1}{c}\int \frac{dx}{\sqrt{1-(1-\frac{\mathcal{E}-U}{mc^2})^2}} \approx \int \frac{dx}{\sqrt{\frac{2}{m}(\mathcal{E}-U)}}$$

in agreement with nonrel. expectations
$-$Can find relat. $\mathcal{O}(\mathcal{E}/mc^2)$ corrections:

$$\Rightarrow t - t_0 \approx \frac{1}{c}\int \frac{dx}{\sqrt{1-(1-\frac{\mathcal{E}-U}{mc^2})^2}} \approx \int \frac{dx}{\sqrt{\frac{2}{m}(\mathcal{E}-U)}}\left(1+\frac{\mathcal{E}-U}{4mc^2}\right) + \ldots$$

Example 2

A constant force $\boldsymbol{F}$ acts on the particle in the lab frame. Write out the lagrangian and the corresponding equations of motion

$$S \qquad = \int dt\, L, \qquad L = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}} + \boldsymbol{F} \cdot \boldsymbol{r}$$

–Potential energy: $U = -\boldsymbol{F} \cdot \boldsymbol{r}$

►$\boldsymbol{F} = \text{const}$ is not Lorentz-covariant since $\boldsymbol{F} \cdot \boldsymbol{r}$ is 3D scalar, not 4-scalar. Also, recall that $\boldsymbol{r}$ is spacial part of vector x^μ, under boosts transforms as

$$\boldsymbol{r}' = \frac{\boldsymbol{r} - \boldsymbol{V}\, t}{\sqrt{1 - V^2/c^2}}$$

–Equation of motion:

$$\frac{d\boldsymbol{p}}{dt} = \boldsymbol{F}$$

Trajectories in 1D: (assume x in direction of $\boldsymbol{F}$, and initial velocity $v_{y,z} = 0$)

$$t - t_0 = \frac{1}{c} \int \frac{dx}{\sqrt{1 - \left(\frac{mc^2}{E - U(x)}\right)^2}} \quad \Rightarrow \quad x(t) = \frac{mc^2}{F} \sqrt{1 + \left(F\left(t - t_0\right)/mc\right)^2}$$

Control question

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}$$

$$\boldsymbol{p} = \frac{m\boldsymbol{v}}{\sqrt{1 - v^2/c^2}}, \quad E = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

Assume that we have a system of N noninteracting particles. How we should generalize the results shown in this slide?

Control question 2

$$S = -\sum_i m_i c^2 \int d\tau_i = -\sum_i m_i c^2 \int dt \sqrt{1 - \frac{v_i^2}{c^2}}$$

$$\boldsymbol{p} = \sum_i \frac{m\boldsymbol{v}_i}{\sqrt{1 - v_i^2/c^2}}, \quad E = \sum_i \frac{m_i c^2}{\sqrt{1 - v_i^2/c^2}}$$

In classical mechanics we know that for ensemble of particles interacting via central forces the total momentum $\sum m\boldsymbol{v}$ is conserved. If we consider relativistic particle, can we expect similar conservation of nonrelativistic momentum

$$\boldsymbol{P}_{\mathrm{CM}} = \sum m_i \boldsymbol{v}_i = \frac{d}{dt} \sum m_i \boldsymbol{r}_i$$

or relativistic momentum

$$\boldsymbol{p} = \sum \frac{m\boldsymbol{v}_i}{\sqrt{1 - v_i^2/c^2}} \qquad ???$$

Relativistic Kepler problem

In Lecture 7 we analyzed the motion of ***nonrelativistic*** particle in central field, and considered in detail motion in Coulomb potential ("Kepler problem"). Repeat this analysis for ***relativistic*** particle of mass m:

– Write out the lagrangian of the system, identify its symmetries and construct all integrals of motion which might exist in this problem.

– Try to integrate the equations of motion and find trajectories for bound orbits. Analyze if in the Coulomb's field a relativistic particle may fall on center, and if all the trajectories are closed for bound orbits.

– Check if the Kepler's laws remain valid for relativistic particle.

– Demonstrate that in the limit $c \to \infty$ all your results reproduce nonrelativistic results of Lecture 7.

Relativistic Kepler Problem

–Lagrangian:

$$L \quad = -mc^2\sqrt{1-\frac{v^2}{c^2}} - U(r) = -mc^2\sqrt{1-\frac{\dot{r}^2 + r^2\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right)}{c^2}} + \frac{\alpha}{r}, \quad \alpha > 0$$

–generalized coordinates: r, θ, ϕ
–generalized momenta:

$$p_r = \frac{\partial L}{\partial \dot{r}} = \gamma m\,\dot{r}, \quad \gamma \equiv \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}, \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = \gamma m\, r^2\dot{\theta},$$

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = \gamma m\, r^2 \sin^2\theta\dot{\phi} = \mathrm{const}$$

–Explicit Euler-Lagrange equations are extremely complicated, so instead will use integrals of motion to find the trajectory
–**Symmetries** & integrals of motion:

$$\phi \quad \to \phi + \phi_0 \Rightarrow p_\phi \equiv M_z = \mathrm{const}, \qquad t \to t + t_0 \Rightarrow E = \mathrm{const}$$

$$E \quad = p_r \cdot \dot{r} + p_\theta \cdot \dot{\theta} + p_\phi \cdot \dot{\phi} - L = mc^2\gamma - \frac{\alpha}{r}, \quad \alpha > 0$$

–In spherical coordinates we don't see other symmetries, but ...

Relativistic Kepler Problem

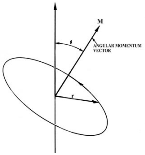
– In Cartesian coordinates we know that potential is invariant w.r.t. rotation around *any* axis

$$\Downarrow$$

All components of $\boldsymbol{M}$ are conserved, $\boldsymbol{M}$ keeps its direction

– In spherical coordinates the orientation of basis vectors $\boldsymbol{e}_\theta, \boldsymbol{e}_\phi$ changes, so we don't see this property (actually there is a hidden symmetry which mixes θ, ϕ in non-trivial way)

Consequences of $\vec{M} = \text{const}$:



- *Particle <u>always</u> moves in a plane ortogonal to $\vec{M}$, i.e. vectors $(\vec{r}(t),\ \vec{v}(t))$ always remain in this plane, for $\forall t$.*

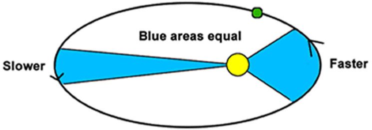
– We may choose plane $\theta = \pi/2$ as a plane of motion, like in nonrelativistic case

$$M \equiv \gamma m r^2 \dot{\phi} = \frac{m r^2 \dot{\phi}}{\sqrt{1 - v^2/c^2}} = \text{const}$$

<u>**Second law of Kepler is NOT valid**</u>

– Due to $\gamma \neq 1$, see that $r^2 d\phi \neq \text{const}$ for relativistic particle in general.

– It works ONLY for special case when particle has $v = \text{const}$ (energy conservation: this is possible only along circular orbit with $r = \text{const}$)



Relativistic Kepler Problem

- Energy conservation (motion plane $\theta = \pi/2$ orthogonal to $\boldsymbol{M}$)

$$E = m\gamma c^2 + U(r),$$

$$\gamma = \left[1 - \frac{v^2}{c^2}\right]^{-1/2} = \left[1 - \frac{\dot{r}^2 + r^2\dot{\phi}^2}{c^2}\right]^{-1/2}$$

$$M_z = m\gamma r^2\dot{\phi} = \mathrm{const}$$

$$\Rightarrow M_z^2\left(1 - \frac{\dot{r}^2 + r^2\dot{\phi}^2}{c^2}\right) = m^2 r^4 \dot{\phi}^2$$

$$\dot{\phi} = M_z\sqrt{\frac{1 - \dot{r}^2/c^2}{m^2r^4 + M_z^2 r^2/c^2}}$$

$$\gamma = \frac{M_z}{mr^2\dot{\phi}} = \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}}$$

$$E = mc^2\sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} - \frac{\alpha}{r}$$

In **nonrelativistic case** ($c \to \infty$) this result yields

$$E_{\mathrm{nonrel}} = E - mc^2 = \qquad (2)$$
$$= m\dot{r}^2 + \frac{M_z^2}{mr^2} - \frac{\alpha}{r} + \mathcal{O}\left(\frac{1}{c^2}\right)$$

and we interpreted the last 2 terms as an effective potential.

– The region $r = 0$ was "protected" by centrifugal forces: for finite energy E, we could not achieve $r = 0$.

Relativistic Kepler Problem

- Energy conservation (motion plane $\theta = \pi/2$ orthogonal to $\boldsymbol{M}$)

$$E = m\gamma c^2 + U(r)\,,$$

$$\gamma = \left[1 - \frac{v^2}{c^2}\right]^{-1/2} = \left[1 - \frac{\dot{r}^2 + r^2\dot{\phi}^2}{c^2}\right]^{-1/2}$$

$$M_z = m\gamma r^2\dot{\phi} = \mathrm{const}$$

$$\Rightarrow M_z^2\left(1 - \frac{\dot{r}^2 + r^2\dot{\phi}^2}{c^2}\right) = m^2 r^4\dot{\phi}^2$$

$$\dot{\phi} = M_z\sqrt{\frac{1 - \dot{r}^2/c^2}{m^2 r^4 + M_z^2 r^2/c^2}}$$

$$\gamma = \frac{M_z}{mr^2\dot{\phi}} = \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}}$$

$$E = mc^2\sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} - \frac{\alpha}{r}$$

In **nonrelativistic case** ($c \to \infty$) this result yields

$$E_{\mathrm{nonrel}} = E - mc^2 = \qquad (2)$$

$$= m\dot{r}^2 + \frac{M_z^2}{mr^2} - \frac{\alpha}{r} + \mathcal{O}\left(\frac{1}{c^2}\right)$$

and we interpreted the last 2 terms as an effective potential.

– The region $r = 0$ was "protected" by centrifugal forces: for finite energy E, we could not achieve $r = 0$.

Relativistic Kepler Problem

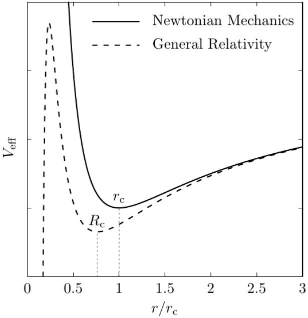
$$E \qquad = mc^2\sqrt{\frac{1+(M_z/mrc)^2}{1-\dot{r}^2/c^2}} - \frac{\alpha}{r} \qquad (1)$$

– We'll try to rewrite (1) as:

$$\frac{mc^2}{2} = \frac{m\dot{r}^2}{2} + \qquad (2)$$
$$+\frac{m^3c^6}{2}\left(E+\frac{\alpha}{r}\right)^{-2}\left(1+\left(\frac{M_z}{mrc}\right)^2\right)$$

– Can interpret the last term as "potential" and use it for qualitative interpretation, however there is some peculiarity:
 * "Potential" depends on true energy E
 * Should consider mc^2 as "energy" (see left-hand side)
 * Large-c expansion near $E \approx mc^2$

$$V_{\text{eff}} \approx -\frac{\alpha}{r} + \frac{M_z^2}{2mr^2} - \frac{\alpha M_z^2}{mc^2}\frac{1}{r^3} + \frac{\alpha^2 M_z^2}{2m^2c^4}\frac{1}{r^4}$$



Solution of EOMs:

$$\frac{dr}{dt} = f(r,E) =$$
$$\sqrt{c^2 - m^2c^6\left(E+\frac{\alpha}{r}\right)^{-2}\left(1+\left(\frac{M_z}{mrc}\right.\right.}$$

$$\int^r \frac{dr}{f(r,E)} = t - t_0 \qquad (3)$$

The integral over r is expressed in terms of elliptic functions.

Relativistic Kepler Problem

Solution of EOMs:

$$\frac{dr}{dt} = f(r,E) = \sqrt{c^2 - m^2 c^6 \left(E + \frac{\alpha}{r}\right)^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right)}, \quad \gamma = \ldots = \frac{1}{mc^2}\left(E + \frac{\alpha}{r}\right)$$

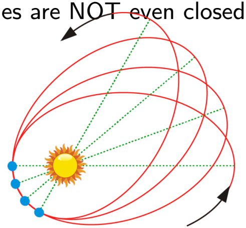
–Equation for trajectory $r(\phi)$:

$$\frac{d\phi}{dr} = \frac{\dot{\phi}}{\dot{r}} = \frac{M_z c^2}{r^2 f(r,\, E)} \left(E + \frac{\alpha}{r}\right) \equiv g(r,E)$$

$$\Rightarrow \phi - \phi_0 = \int dr\, g(r,E)$$

*Trajectories are NOT conical sections, the 1st and 3rd Kepler's laws are NOT valid.

*Trajectories are NOT even closed (precession for Mercury orbit: 14.3''/century).



Motion in external vector (=electromagnetic) field

Landau: Minimal action for a free particle:

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}$$

postulate that for real trajectory $S = \min$.

▶ Now we will consider motion in external 4-vector field $A_\mu(x)$

−We'll see that this reproduces equations for electromagentism. Our goal is just to learn how to write lagrangian covariantly, so we will not discuss any experimental aspects/foundations.

▶We assume that interaction might be introduced by additional interaction term $S_{\rm int}$ in our action

▶Result should be covariant (the same in all frames)

Motion in external EM field

▷Let's assume

$$S_{\text{int}} = -q \int dx_\mu A^\mu(x) = q \int d\tau \, (u \cdot A) \,,$$

$$u_\mu \equiv \frac{dx_\mu}{d\tau}; \quad u \cdot A \equiv \eta_{\mu\nu} u^\mu A^\nu$$

If we use notation $A^\mu = (\phi, \boldsymbol{A})$, then

$$S_{\text{int}} = -q \int d\tau \, \frac{\phi - \boldsymbol{A} \cdot \boldsymbol{v}}{\sqrt{1 - v^2/c^2}} = -q \int dt \, (\phi - \boldsymbol{A} \cdot \boldsymbol{v})$$

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - q \, (\phi - \boldsymbol{A} \cdot \boldsymbol{v})$$

Write out the Equations of Motion for such lagrangian and demonstrate that it agrees with Lorentz formula.

$$\frac{d\boldsymbol{p}}{dt} = q \, (\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$$

Motion in external EM field

$$L = -mc^2\sqrt{1-\frac{v^2}{c^2}} - q\left(\phi - \boldsymbol{A}\cdot\boldsymbol{v}\right)$$

► Generalized momentum:

$$P^a = \frac{\partial L}{\partial v^a} = \underbrace{\frac{mv^a}{\sqrt{1-\frac{v^2}{c^2}}}}_{p^a} + q\,A^a$$

► p^a-momentum of free particle; $P^a \neq p^a$

► Equation of motion:

$$\frac{dP^a}{dt} \equiv \frac{dp^a}{dt} + q\frac{dA^a(x,\,t)}{dt} = \frac{\partial L}{\partial r^a} = -q\,\nabla_a\phi + q\,\nabla_a\left(\boldsymbol{A}\cdot\boldsymbol{v}\right)$$

Motion in external EM field

$$\frac{dP^a}{dt} \equiv \frac{dp^a}{dt} + q\frac{dA^a(x,\,t)}{dt} = \frac{\partial L}{\partial r^a} = -q\,\nabla_a\phi + q\,\nabla_a\left(\boldsymbol{A}\cdot\boldsymbol{v}\right)$$

►Claim:

$$\boldsymbol{\nabla}\left(\boldsymbol{A}\cdot\boldsymbol{v}\right) \equiv \boldsymbol{v}\times\left[\nabla\times\boldsymbol{A}\right] + \left(\boldsymbol{v}\cdot\nabla\right)\boldsymbol{A}$$

$(\boldsymbol{a}\times[\boldsymbol{b}\times\boldsymbol{c}] = \boldsymbol{b}\,(\boldsymbol{a}\cdot\boldsymbol{c}) - \boldsymbol{c}\,(\boldsymbol{a}\cdot\boldsymbol{b})$, so $\boldsymbol{b}\,(\boldsymbol{a}\cdot\boldsymbol{c}) = \boldsymbol{a}\times[\boldsymbol{b}\times\boldsymbol{c}] + \boldsymbol{c}\,(\boldsymbol{a}\cdot\boldsymbol{b}))$

►Claim:

$$\frac{d\boldsymbol{A}(x,\,t)}{dt} \equiv \frac{\partial\boldsymbol{A}(x,\,t)}{\partial t} + \frac{dx^b}{dt}\,\frac{\partial\boldsymbol{A}(x,\,t)}{\partial x^b} = \left[\frac{\partial\boldsymbol{A}(x,\,t)}{\partial t} + \left(\boldsymbol{v}\cdot\nabla\right)\boldsymbol{A}\right]$$

so

$$\begin{aligned} \frac{d\boldsymbol{p}}{dt} &= q\left(\boldsymbol{E} + \boldsymbol{v}\times\boldsymbol{B}\right) \\ \boldsymbol{E} &= -\boldsymbol{\nabla}\phi - \frac{\partial\boldsymbol{A}}{\partial t}, \quad \boldsymbol{B} = \boldsymbol{\nabla}\times\boldsymbol{A} \end{aligned}$$

►Magnetic field does not change kinetic energy:

$$\frac{dE_{\mathrm{kin}}}{dt} = \boldsymbol{v}\cdot\frac{d\boldsymbol{p}}{dt} = q\,\boldsymbol{E}\cdot\boldsymbol{v}$$

Motion in external electromagnetic field

►Explicit derivation of energy E from the Noether's theorem:s

$$E = \sum_a \frac{\partial L}{\partial v_a} v_a - L = \mathrm{const}$$

$$\boldsymbol{P} = \frac{\partial L}{\partial \boldsymbol{v}} \equiv \boldsymbol{p} + q\,\boldsymbol{A}$$

$$\begin{aligned} E \qquad &= \boldsymbol{v}\cdot\boldsymbol{P} - L = \frac{mv^2}{\sqrt{1-v^2/c^2}} + \cancel{q\,\boldsymbol{A}\cdot\boldsymbol{v}} + mc^2\sqrt{1-\frac{v^2}{c^2}} + q\left(\phi - \cancel{\vec{\boldsymbol{A}}\cdot\vec{\boldsymbol{v}}}\right) \\ &= \frac{mc^2}{\sqrt{1-v^2/c^2}} + q\,\phi \end{aligned}$$

−Magnetic potential $\boldsymbol{A}$ does not contribute to energy, in agreement with our expectations

Motion in external EM field (covariant)

[Sec. 7.10 in *Goldstein*]

► So far we demonstrated that the action

$$S = S_0 + S_{\rm int}$$
$$S_0 = -mc^2 \int d\tau, \quad S_{\rm int} = q \int d\tau \, (u \cdot A)$$

correctly reproduced Lorentz force, yet we found that the expression is NOT Lorentz-covariant.
– *Reason*: We explicitly separated time t and coordinates $\boldsymbol{x}$

► **Our goal**: get covariant form of equations of motion
– *Method*:
*We'll assume the trajectory is given by $x^\mu(\tau)$ and NOT separating explicitly coordinates and time in lab-frame.

*We'll analyze change of action δS in case of variation of the trajectory

$$x^\mu(\tau) \to x^\mu(\tau) + \delta x^\mu(\tau)$$

and after that request $\delta S/\delta x^\mu = 0$.

Motion in external EM field (covariant)

► Variation of integrand of S_0:

$$
\begin{aligned}
d\tau &\equiv \sqrt{\eta_{\mu\nu} dx^\mu dx^\nu} \\
\delta\, d\tau &= \frac{\eta_{\mu\nu} dx^\mu d\delta x^\nu}{\sqrt{\eta_{\alpha\beta} dx^\alpha dx^\beta}} = \eta_{\mu\nu} \frac{dx^\mu}{d\tau} d\delta x^\nu = \eta_{\mu\nu} u^\mu d\delta x^\nu \equiv u_\nu d\delta x^\nu
\end{aligned}
$$

► Variation of integrand of $S_{\rm int}$

$$
\delta\left(dx_\mu A^\mu(x)\right) = \left(d\delta x_\mu\right) A^\mu(x) + \left(dx_\mu\right) \frac{\partial A^\mu(x)}{\partial x^\nu} \delta x^\nu
$$

► For the terms $\int d\tau\, f_\mu(\ldots) d\delta x^\mu$ we use integration by parts

$$
f_\mu(x) d\delta x^\mu \equiv d\left(f_\mu(x)\delta x^\mu\right) - df_\mu(\ldots)\delta x^\mu = d\left(f_\mu(x)\delta x^\mu\right) - \frac{\partial f_\mu(\ldots)}{\partial x^\nu} dx^\nu\, \delta x^\mu
$$

$$
\int_{\tau_1}^{\tau_2} d\tau\, f_\mu(\ldots) d\delta x^\mu = f_\mu(x)\delta x^\mu\Big|_{\tau_1}^{\tau_2} - \int \frac{\partial f_\mu(x)}{\partial x^\nu} dx^\nu\, \delta x^\mu = -\int d\tau\, \frac{\partial f_\mu(x)}{\partial x^\nu} u^\nu\, \delta x^\mu
$$

Motion in external EM field (covariant)

► Variaton of action:

$$\delta S = \int d\tau\, \delta x^\mu \left[mc^2 \frac{du^\nu}{d\tau} - q\, F^{\mu\alpha} u_\alpha \right] = 0,$$

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$m\,\frac{du^\nu}{d\tau} \equiv \frac{dp^\nu}{d\tau} = q\, F^{\mu\nu} u_\nu \qquad (1)$$

► $F_{\mu\nu}$ is called **electromagnetic field tensor**

Write out explicitly all the components of $F_{\mu\nu}$ and demonstrate that (1) is equivalent to

$$\begin{aligned} \frac{d\boldsymbol{p}}{dt} &= q\,(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) \\ \boldsymbol{E} &= -\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t}, \\ \boldsymbol{B} &= \boldsymbol{\nabla} \times \boldsymbol{A} \end{aligned}$$

(assume $A^\mu = (\phi/c,\, \boldsymbol{A})$, $A_\mu = \eta_{\mu\nu} A^\nu = (\phi/c,\, -\boldsymbol{A})$)

Motion in external EM field (covariant)

$$d\tau = dt\,\sqrt{1 - v^2/c^2}$$

$$m\,\frac{du^\mu}{d\tau} \equiv \frac{dp^\mu}{d\tau} = q\,F^{\mu\nu}u_\nu \qquad (1)$$

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$u^0 = u_0 = \frac{c}{\sqrt{1 - v^2/c^2}}$$

$$u^j = -u_j = \frac{v^j}{\sqrt{1 - v^2/c^2}},$$

▶Explicit form:

$$F^{\mu\nu} = \begin{bmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

-factors $1/\sqrt{1 - v^2/c^2}$ appear in both parts and cancel

▷ Eq. (1) in components ($p_0 \equiv E_{\rm kin}/c$):

$$F_{\mu\nu} = \eta_{\mu\alpha}F^{\alpha\beta}\eta_{\beta\nu} = \begin{bmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

$$\frac{dp_0}{dt} = \frac{q}{c}\,(E_x v_x + E_y v_y + E_z v_z) = \frac{q}{c}\boldsymbol{E}\cdot$$

$$\frac{dp_x}{dt} = q\,(E_x + v_y B_z - v_z B_y)\,,$$

$$\frac{dp_y}{dt} = q\,(E_y + v_z B_y - v_y B_z)\,,$$

$$\frac{dp_z}{dt} = q\,(E_z + v_x B_y - v_y B_x)\,,$$

▷Use $\eta_{\mu\nu}$, $\eta^{\mu\nu}$ to raise/lower indices

-signs in front of electric field change when raising/lowering indices, so be careful

$$E_i = c\,F_{0i},\;\; B_i = -\frac{1}{2}\epsilon_{ijk}F^{jk}$$

in agreement with results which we got in lab-frame

Control question

−In our derivation we *assumed* that interaction term is linear in A^μ,

$$S_{\rm int} = -q\int dx_\mu A^\mu(x) = q\int d\tau\,(u\cdot A)\,,$$

$$u_\mu \equiv \frac{dx_\mu}{d\tau};\quad u\cdot A \equiv \eta_{\mu\nu}u^\mu A^\nu$$

What happens if we include higher-order terms, e.g.

$$S^{\rm (extra)}_{\rm int} = \lambda_2\int d\tau\;(u\cdot A)^2 + \lambda_3\int d\tau\;(u\cdot A)^3$$

−*Are these terms compatible with Lorentz-covariance of the action?*
−*Why physically these terms should not be included for electromagnetism?*

Lagrangian for a system of N particles

Write out the lagrangian of a system which consists of N electric charges moving with relativistic velocities

Lagrangian for a system of N particles

►**We cannot use long-range forces, contradicts relativity**

– ... instead should consider as a system N particles + EM field

$$S \qquad = S_0 + S_{\rm int} + \int d^2x\, \mathcal{L}_{\rm field}$$

►Interaction always has local form. Nonlocal terms would contradict special relativity

►Should include term $\mathcal{L}_{\rm field}$ which describes the dynamics of the field itself

– Lagrangian $\mathcal{L}_{\rm field}$ can be constructed only from $F_{\mu\nu}$ (gauge invariant)

Courses of electromagnetism:

1) The only possible construction:

$$\mathcal{L}_{\rm field} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

2) Variation of A_μ yields extra equations for electromagnetic field (Maxwell's equations):

$$\partial_\nu F^{\mu\nu} = \mu_0 j^\mu \quad \epsilon_{\alpha\beta\mu\nu}\partial^\beta F^{\mu\nu} = 0$$

Motion in external gravitational field

Landau: Minimal action for a free particle:

$$S = -mc^2 \int d\tau = -mc^2 \int dt \sqrt{1 - \frac{v^2}{c^2}}$$

postulate that for real trajectory $S = \min$.

- ▶ Up to now we always assumed that the interaction might be introduced by additional interaction term $S_{\rm int}$
- ▶ Adding $S_{\rm int}$ is not the only covariant way to introduce interaction. For example, we could replace flat metrics

$$\eta_{\mu\nu} \qquad = \mathrm{diag}(1, -1, -1, -1)$$

as $\eta_{\mu\nu} \to g_{\mu\nu}(x)$, where matrix $g_{\mu\nu}(x)$ called "metric tensor".

– We'll also use notation $g^{\mu\nu}(x)$ for the inverse

– Eventually this leads to General Relativity (GR, theory of gravity)

–... complicated topic, we will only touch it to learn how to write covariantly the equations of motion. We will omit all the experimental evidence/checks

– Our discussion will be mostly introductory and based on [Chapter 7 in *Goldstein*]

– ... for more detailed discussion see [Landau, Vol. 2] (though we will not go that far, there is a dedicated course on this)

Motion in external gravitational field

Landau: Minimal action for a particle:

$$S = -mc^2 \int d\tau \sqrt{g_{\mu\nu}(x) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}}$$

$-\tau$ could be any parameter along the particle's trajectory. If it is proper time, then $\sqrt{g_{\mu\nu}\ldots} = 1$

- As usual assume the variation vanishes at the endpoints
- Einstein's convention implied (summation over dummy repetitive indices)
- Variation of particle's trajectory:

$$\begin{aligned}
&-\delta S/mc^2 = \\
&= \int d\tau \frac{\partial g_{\mu\nu}(x)}{\partial x^\lambda} \delta x^\lambda \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + \int d\tau\, g_{\mu\nu}(x) \frac{d\delta x^\mu}{d\tau} \frac{dx^\nu}{d\tau} + \int d\tau\, g_{\mu\nu}(x) \frac{dx^\mu}{d\tau} \frac{d\delta x^\nu}{d\tau} \\
&= \int d\tau\, \delta x^\lambda \left(\frac{\partial g_{\mu\nu}(x)}{\partial x^\lambda} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} - \frac{\partial}{\partial \tau} \left(g_{\mu\lambda}(x) \frac{dx^\mu}{d\tau} \right) - \frac{\partial}{\partial \tau} \left(g_{\lambda\mu}(x) \frac{dx^\mu}{d\tau} \right) \right)
\end{aligned}$$

$$\frac{\partial g_{\mu\lambda}(x)}{\partial \tau} \to \frac{\partial x^\nu}{\partial \tau} \frac{\partial g_{\mu\lambda}(x)}{\partial x^\nu}$$

$$0 = \int d\tau \delta x^\mu \left(-2 g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} + \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\mu g_{\alpha\nu} - 2 \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\alpha g_{\mu\nu} \right)$$

Motion in external gravitational field

$$0 = \int d\tau \delta x^\mu \left(-2g_{\mu\nu} \frac{d^2x^\nu}{d\tau^2} + \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\mu g_{\alpha\nu} - 2 \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \partial_\alpha g_{\mu\nu} \right)$$

$$g_{\mu\nu} \frac{d^2x^\nu}{d\tau^2} + \frac{1}{2} \frac{dx^\alpha}{d\tau} \frac{dx^\nu}{d\tau} \left(\partial_\alpha g_{\mu\nu} + \partial_\nu g_{\mu\alpha} - \partial_\mu g_{\alpha\nu} \right) = 0$$

- "Equations of motion" (geodesics):

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0$$
$$\Gamma^\mu_{\alpha\beta} = \frac{g^{\mu\nu}}{2} \left(\frac{\partial g_{\alpha\nu}}{\partial x^\beta} + \frac{\partial g_{\nu\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\nu} \right)$$

- Symbols $\Gamma$are called Cristoffel's symbols
- We may rewrite

$$\frac{dp^\mu}{d\tau} = -m\, \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau}$$

so the right-hand side might be interpreted as a covariant 4-force which is proportional to mass m (recall equality of gravitational and inertial masses)